

14. For primitive statements p, q ,

a) verify that $p \rightarrow [q \rightarrow (p \wedge q)]$ is a tautology.

b) verify that $(p \vee q) \rightarrow [q \rightarrow q]$ is a tautology

c) is $(p \vee q) \rightarrow [q \rightarrow (p \wedge q)]$ a tautology?

Sol

Q) We have :

$$\Rightarrow p \rightarrow [q \rightarrow (p \wedge q)]$$

$\Rightarrow$ only 2 possible cases

Case (1) : p is True

$$\Rightarrow T \rightarrow [q \rightarrow (T \wedge q)]$$

$$\Rightarrow q \rightarrow (T \wedge q)$$

$$\Rightarrow q \rightarrow q$$

$$\Rightarrow \neg q \vee q$$

$\Rightarrow$ True always

Case (2) : P is false

$$\Rightarrow F \rightarrow [q \rightarrow (F \wedge q)]$$

$$\Rightarrow F \rightarrow [q \rightarrow F]$$

$$\Rightarrow F \rightarrow T$$

$\Rightarrow$ True Always

$\therefore$ given prop. is a tautology

b) given: $(p \vee q) \rightarrow [q \rightarrow q]$

$$: (p \vee q) \rightarrow T$$

: True Always

$\therefore$ Tautology

c) given: $(p \vee q) \rightarrow [q \rightarrow (p \wedge q)]$

Case ① p is False

$$\rightarrow (F \vee q) \rightarrow [q \rightarrow (F \wedge q)]$$

$$\Rightarrow q \rightarrow [q \rightarrow F]$$

$$\Rightarrow q \rightarrow \neg q$$

$$\Rightarrow \neg q \vee \neg q$$

$\Rightarrow \underline{\neg q}$ $\therefore$ contingency
not
tautology

